

① Find the sum for each of the following series:

$$\begin{aligned}
 a) \sum_{n=1}^{\infty} \frac{2}{3^n} &= \frac{2}{3} + \frac{2}{3^2} + \frac{2}{3^3} + \dots + \frac{2}{3^n} + \dots \\
 &= 2 \left( \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots + \frac{1}{3^n} + \dots \right) \\
 &= 2 \cdot \frac{1}{3} \left( 1 + \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-1}} + \frac{1}{3^n} + \dots \right) \\
 &= \frac{2}{3} \cdot \lim_n \frac{\frac{1}{3}^{n+1} - 1}{\frac{1}{3} - 1} \\
 &= \frac{2}{3} \cdot \frac{0 - 1}{-\frac{2}{3}} \\
 &= \frac{2}{3} \cdot \frac{3}{2} \\
 &= 1.
 \end{aligned}$$

$$\sum_{n=0}^{\infty} 2^n = 1 + 2 + 2^2 + \dots + 2^n + \dots = \lim_n \frac{2^{n+1} - 1}{2 - 1} = \frac{-1}{2 - 1} = \frac{1}{1 - 2}.$$

$|2| < 1$ .

$$\begin{aligned}
 b) \sum_{n \geq 1} \frac{2^{n+1}}{n!} &= \sum_{n=1}^{\infty} \left( \frac{2^n}{n!} + \frac{1}{n!} \right) = \sum_{n=1}^{\infty} \frac{2^n}{(n-1)!} + \sum_{n=1}^{\infty} \frac{1}{n!} = \\
 &= 2 \cdot \sum_{n=0}^{\infty} \frac{1}{n!} + \sum_{n=1}^{\infty} \frac{1}{n!} = 2 \cdot e + e - 1 = 3e - 1.
 \end{aligned}$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e \rightarrow \sum_{n=1}^{\infty} \frac{1}{n!} = e - 1$$

$$\sum_{n=1}^{\infty} \frac{1}{(n-1)!} = \sum_{n=0}^{\infty} \frac{1}{n!}$$

$$c) \sum_{n \geq 1} \frac{1}{4^{n^2-1}} = \sum_{n \geq 1} \frac{1}{(2n+1)(2n-1)} = \sum_{n \geq 1} \frac{1}{(2n+1)(2n-1)} =$$

$$= \frac{1}{2} \sum_{n \geq 1} \frac{1}{(2n+1)(2n-1)}$$

$$= \frac{1}{2} \lim_n \left( \frac{1}{1} - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \dots + \frac{1}{2n-1} - \frac{1}{2n+1} \right)$$

$$= \frac{1}{2} \lim_n \left( 1 - \underbrace{\frac{1}{2^{n+1}}}_{\rightarrow 0} \right) = \frac{1}{2} \cdot 1 = \frac{1}{2}.$$

$$\begin{aligned} d) \sum_{n \geq 1} \frac{1}{n(n+1)(n+2)} &= \sum_{n \geq 1} \frac{1}{2} \cdot \frac{n+2-n}{n(n+1)(n+2)} = \frac{1}{2} \sum_{n \geq 1} \frac{2}{n(n+1)(n+2)} = \\ &= \frac{1}{2} \cdot \sum_{n \geq 1} \frac{n+2-n}{n(n+1)(n+2)} = \frac{1}{2} \sum_{n \geq 1} \left( \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right) \\ &= \frac{1}{2} \cdot \sum_{n=1}^{\infty} \frac{1}{n(n+1)} - \frac{1}{2} \cdot \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} \\ &= \frac{1}{2} \cdot \sum_{n=1}^{\infty} \frac{1}{n(n+1)} - \frac{1}{2} \cdot \sum_{n=2}^{\infty} \frac{1}{n(n+1)} \\ &= \frac{1}{2} \cdot 1 - \frac{1}{2} \cdot 1 \\ &= \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{n(n+1)} &= \sum_{n=1}^{\infty} \frac{n+1-n}{n(n+1)} = \\ &= \sum_{n=1}^{\infty} \left( \frac{1}{n} - \frac{1}{n+1} \right) \\ &= \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{n} - \frac{1}{n+1} \\ &\rightarrow \lim_n \left( 1 - \frac{1}{n+1} \right) = 1. \end{aligned}$$

$$\begin{aligned} e) S &= \sum_{n=1}^{\infty} \frac{n}{2^n} = \frac{1}{2^1} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} + \dots \\ &= \frac{1}{2} + \frac{1}{2} \left( \frac{2}{2} + \frac{3}{2^2} + \dots + \frac{n}{2^{n-1}} + \dots \right) \\ &= \frac{1}{2} + \frac{1}{2} \left( \frac{2}{2} + \frac{3}{2^2} + \dots + \frac{n-1+1}{2^{n-1}} + \dots \right) \\ &= \frac{1}{2} + \frac{1}{2} \cdot \left( \frac{2-1+1}{2^{2-1}} + \frac{3-1+1}{2^{3-1}} + \dots + \frac{n-1+1}{2^{n-1}} + \dots \right) \\ &= \frac{1}{2} + \frac{1}{2} \cdot \left( \underbrace{\frac{1}{2} + \frac{2}{2^2} + \dots + \frac{n-1}{2^{n-1}}}_S + \underbrace{\frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}}}_1 \right) \\ &= \frac{1}{2} + \frac{1}{2} \cdot (S+1) \end{aligned}$$

$$\Rightarrow S = \frac{1}{2} + \frac{1}{2} (S+1) \quad | \cdot 2 \Rightarrow 2S = 1 + S + 1 \Rightarrow 2S = S + 2 \Rightarrow S = 2$$

$$\sum_{n=0}^{\infty} \dots = \frac{1}{1-q} = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots = \frac{1}{1-\frac{1}{2}} = 2.$$

$$\sum_{n=1}^{\infty} = 2 \text{ also } 2??$$

Method 2:  $S = \sum_{n=1}^{\infty} \frac{n}{2^n} = \sum_{n=1}^{\infty} n \cdot \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \underbrace{n \cdot x^n}_{x \cdot (x^n)'}, x = \frac{1}{2}$

$$S = \sum_{n=1}^{\infty} x \cdot (x^n)' = x \cdot \sum_{n=1}^{\infty} (x^n)', x = \frac{1}{2}$$

$$(f+g)' = f' + g'$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - f \cdot g'}{g^2}$$

$$\sum_{n=1}^{\infty} (x^n)' = \lim_{N \rightarrow \infty} \sum_{n=1}^N (x^n)' = \lim_N \left( \sum_{n=1}^N x^n \right)'$$

$$\left( \sum_{n=1}^N x^n \right)' = \left( \frac{x^{N+1} - 1}{x - 1} - 1 \right)' = \frac{(N+1)x^N \cdot (x-1) - (x^{N+1} - 1) \cdot 1}{(x-1)^2} =$$

$$= \frac{(N+1)(x^{N+1} - x^N) - x^{N+1} + 1}{(x-1)^2} = \frac{(N+1)x^{N+1} - (N+1)x^N - x^{N+1} + 1}{(x-1)^2} =$$

$$\stackrel{0}{=} \frac{N \cdot x^{N+1} - (N+1)x^N + 1}{(x-1)^2}$$

$$= \frac{1}{(x-1)^2}$$

$$\lim_N N \cdot x^{N+1} \stackrel{\left(\frac{1}{2}\right)^N \rightarrow 0}{=} 0, \text{ if } |x| = \frac{1}{2} < 1.$$

$$\lim_N \left( \sum_{n=1}^N x^n \right)' = \lim_N \left( \frac{1}{(x-1)^2} \right) = \frac{1}{(x-1)^2}$$

$$S = x \cdot \frac{1}{(x-1)^2} = \frac{x}{(x-1)^2} \stackrel{x=\frac{1}{2}}{=} \frac{\frac{1}{2}}{\left(\frac{1}{2}-1\right)^2} = \frac{\frac{1}{2}}{\left(-\frac{1}{2}\right)^2} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2$$

f)  $\sum_{n=1}^{\infty} \frac{1}{3\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{3}}} > \sum_{n=1}^{\infty} \frac{1}{n} = \infty$  divergent

$$\sum_{n=1}^{\infty} \frac{x^n}{n} \text{ sum of integrals}$$

$$\sqrt[3]{n} < n \\ \Rightarrow \frac{1}{\sqrt[3]{n}} > \frac{1}{n}$$

② Study the convergence:

a)  $\sum_{n \geq 2} \frac{1}{\ln n}$

Ratio test:  $\frac{x_{n+1}}{x_n} = \frac{\frac{1}{\ln(n+1)}}{\frac{1}{\ln n}} = \frac{\ln n}{\ln(n+1)} = \frac{\ln n}{\ln\left(n\left(1+\frac{1}{n}\right)\right)} =$   
 $= \frac{\ln n}{\ln n + \underbrace{\ln\left(1+\frac{1}{n}\right)}_{\ln 1 = 0}} = \frac{\ln n}{\ln n} = 1$ , inconclusive.

INEQ:

$$\ln n < n. \\ \Rightarrow \frac{1}{\ln n} > \frac{1}{n} \Rightarrow \sum_{n \geq 2} \frac{1}{\ln n} > \sum_{n \geq 2} \frac{1}{n} = \infty \text{ divergent.}$$

b)  $\sum_{n \geq 1} \frac{1}{n\sqrt{n+1}}$

M1 compare with  $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}} > 1$  convergent.

$$\sqrt{n+1} > \sqrt{n} \quad | \cdot n$$

$$n\sqrt{n+1} > n\sqrt{n} \quad |^{-1}$$

$$\frac{1}{n\sqrt{n+1}} < \frac{1}{n\sqrt{n}} \quad | \sum$$

$$\sum_{n \geq 1} \frac{1}{n\sqrt{n+1}} < \sum_{n \geq 1} \frac{1}{n\sqrt{n}} \text{ convergent}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \begin{cases} \text{conv if } p > 1 \\ \text{div if } p \leq 1. \end{cases}$$

M2 Ratio test:  $\lim_n \frac{\frac{1}{n\sqrt{n+1}}}{\frac{1}{n\sqrt{n}}} = \lim_n \frac{\frac{1}{n\sqrt{n}}}{\frac{1}{n\sqrt{n+1}}} = \lim_n \frac{\sqrt{n}}{\sqrt{n+1}} = 1 \in (0, \infty)$

$\Rightarrow \sum_{n \geq 1} \frac{1}{n\sqrt{n+1}}$  has the same nature as  $\sum_{n \geq 1} \frac{1}{n\sqrt{n}}$  (y\_n) convergent

$\Rightarrow x_n = \sum_{n \geq 1} \frac{1}{n\sqrt{n+1}}$  is also convergent.

$\sum \ln\left(1+\frac{1}{n}\right)$  has the same nature as  $\sum \frac{1}{n}$  convergent.

c)  $\sum_{n \geq 1} \frac{1}{n}$  has the same nature as  $\sum_{n \geq 1} \frac{1}{n^2}$

$$\lim_n \frac{x_n}{y_n} = \lim_n \frac{\frac{\ln(1 + \frac{1}{n})}{n}}{\frac{1}{n^2}} = \lim_n n \cdot \ln(1 + \frac{1}{n}) = \lim_n \ln(1 + \frac{1}{n})^n \stackrel{(*)}{=} 1$$

$$\lim_n \ln(1 + \frac{1}{n})^n \stackrel{1^\infty}{=} \ln \lim_n (1 + \frac{1}{n})^n = \ln \lim_n \left[ \left(1 + \frac{1}{n}\right)^n \right]^{\frac{1}{n}} = \ln e = 1 \quad (*)$$

$$\boxed{\begin{aligned} \lim_n n \cdot \ln(1 + \frac{1}{n}) &= \ln e = 1. \\ \Rightarrow \lim_n \frac{\ln(1 + \frac{1}{n})}{\frac{1}{n}} &= 1. \end{aligned}}$$

$$d) \sum_{n=1}^{\infty} \frac{n!}{(n+1)^n} = \sum_{n=1}^{\infty} x_n.$$

$$\text{Ratio test: } \frac{x_{n+1}}{x_n} = \frac{(n+1)!}{(n+2)^{n+1}} \cdot \frac{(n+1)^n}{n!} = \frac{(n+1)^{n+1}}{(n+2)^{n+1}} = \left(\frac{n+1}{n+2}\right)^{n+1}$$

$$\lim_n \frac{x_{n+1}}{x_n} = \lim_n \left(\frac{n+1}{n+2}\right)^{n+1} = \lim_n \left(1 - \frac{1}{n+2}\right)^{n+1} =$$

$$= \lim_n \left[ \left(1 - \frac{1}{n+2}\right)^{-(n+2)} \right]^{\frac{n+1}{-(n+2)}} = e^{-1} = \frac{1}{e} < 1 \Rightarrow \sum x_n \text{ convergent.}$$

$$e) \sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2} = \sum_{n=1}^{\infty} x_n.$$

$$\text{Root test: } \sqrt[n]{x_n} = \left(\frac{n}{n+1}\right)^{n^2 \cdot \frac{1}{n}} = \left(\frac{n}{n+1}\right)^n$$

$$\lim_n \sqrt[n]{x_n} = \lim_n \left(\frac{n+1-1}{n+1}\right)^n = \lim_n \left(1 - \frac{1}{n+1}\right)^n =$$

$$= \lim_n \left[ \left(1 - \frac{1}{n+1}\right)^{-(n+1)} \right]^{\frac{n}{-(n+1)}} = e^{-1} = \frac{1}{e} < 1 \Rightarrow \sum x_n \text{ conv.}$$

$$f) \sum_{n=2}^{\infty} \frac{1}{n \cdot \ln n}$$

$$\text{Ratio test: } \frac{x_{n+1}}{x_n} = \frac{(n \cdot \ln n)}{(n+1) \cdot \ln(n+1)} = 1 \cdot 1 = 1 \text{ inconclusive.}$$

### Cauchy condensation test:

$$\sum_{n=1}^{\infty} x_n \text{ "like" } \sum_{n=1}^{\infty} \frac{2^n \cdot x_{2^n}}{2^n} \quad (\text{have the same nature})$$

$$\sum_{n=2}^{\infty} \frac{1}{n \ln n} \text{ has the same nature as } \sum_{n=1}^{\infty} \frac{1}{2^n \ln 2^n} =$$

$$= \sum_{n=1}^{\infty} \frac{1}{n \ln 2} = \frac{1}{\ln 2} \sum_{n=1}^{\infty} \frac{1}{n} = \infty \text{ divergent.}$$

### Ratio test vs Raabe

$$\begin{aligned} &\rightarrow > 1 \Rightarrow \text{div} \\ &\rightarrow < 1 \Rightarrow \text{conv.} \end{aligned}$$

$$\begin{aligned} &\rightarrow > 1 \Rightarrow \text{conv} \\ &\rightarrow < 1 \Rightarrow \text{div.} \end{aligned}$$

③ a)  $\sum_{n \geq 1} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n} = \sum_{n \geq 1} x_n.$

Ratio:  $\frac{1 \cdot 3 \cdot \dots \cdot (2n+1)}{2 \cdot 4 \cdot \dots \cdot 2n+2} \cdot \frac{2 \cdot 4 \cdot \dots \cdot 2n}{1 \cdot 3 \cdot \dots \cdot 2n-1} = \frac{2n+1}{2n+2} = 1. \text{ inconcl.}$

Raabe-Duhamel:  $\lim_n n \left( \frac{x_n}{x_{n+1}} - 1 \right) =$

$$= \lim_n n \cdot \left( \frac{2n+2}{2n+1} - 1 \right)$$

$$= \lim_n n \cdot \frac{1}{2n+1}$$

$$= \frac{1}{2} < 1 \Rightarrow \sum x_n \text{ divergent.}$$

b)  $\sum_{n \geq 1} a^{\ln n}, a > 0$

Ratio:  $\frac{x_{n+1}}{x_n} = \frac{a^{\ln(n+1)}}{a^{\ln n}} = a^{\ln(n+1) - \ln n} = a^{\ln \frac{n+1}{n}}$

$\lim_n \frac{x_{n+1}}{x_n} = \lim_n a^{\ln \left( \frac{n+1}{n} \right)} \rightarrow 1 = a^0 = 1. \text{ inconcl.}$

Raabe-Duhamel:

$$\lim_n n \cdot \left( \frac{x_n}{x_{n+1}} - 1 \right) = \lim_n n \cdot \left( a^{\ln \frac{n}{n+1}} - 1 \right)$$

$$= \lim_n n \cdot \frac{a^{\ln \frac{n}{n+1}} - 1}{\ln \frac{n}{n+1}} \cdot \ln \frac{n}{n+1}$$

$$= \lim_n \underbrace{\frac{a^{\ln \frac{n}{n+1}} - 1}{\ln \frac{n}{n+1}}}_{\rightarrow \ln a} \cdot \underbrace{\left( \ln \frac{n}{n+1} \right)^n}_{\rightarrow}$$

(1)  $\ln a \cdot (-1) = -\ln a = \ln \frac{1}{a}$

(2)  $\begin{cases} \text{If } \ln \frac{1}{a} < 1 \Rightarrow \text{div.} \\ \text{If } \ln \frac{1}{a} > 1 \Rightarrow \text{conv.} \\ \text{If } \ln \frac{1}{a} = 1 = \ln e \\ \Rightarrow \frac{1}{a} = e \end{cases}$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a(1)$$

$$\lim_n \left( \ln \frac{n}{n+1} \right)^n = \lim_n \left( \ln \frac{n+1-1}{n+1} \right)^n = \lim_n \ln \left( 1 - \frac{1}{n+1} \right)^n =$$

$$= \ln \lim_n \left[ \left( 1 - \frac{1}{n+1} \right)^{-(n+1)} \right]^{-\frac{n}{n+1}} = \ln e^{-1} = -1. (2)$$

$$\Rightarrow \sum a^{\ln n} = \sum \left( \frac{1}{e} \right)^{\ln n}$$

$$= \sum \frac{1}{n} = \infty$$

↓  
div.

$$\boxed{a^{\ln n} = n^{\ln a}} = \frac{1}{n^{-\ln a}} = \frac{1}{n^{\ln \frac{1}{a}}} = p$$

$$\sum \frac{1}{n^p} \begin{cases} \rightarrow p > 1 \Rightarrow \text{conv} \\ \rightarrow p \leq 1 \Rightarrow \text{div.} \end{cases}$$